LAN, ETHERNET & WIRELESS NETWORK INVESTIGATION

Question 1: Explain the responsibilities of the Data-Link Layer and distinguish Logical Link Control from Media Access Control.

Responsibilities of the Data-Link Layer :
The Data-Link Layer is the second layer of the OSI model. It provides node-to-node delivery of data over a particular network link.

Its main responsibilities are:

1. Framing: It divides network-layer packets into frames by adding headers and trailers.
2. Physical addressing: It uses MAC addresses to identify devices on a local network.
3. Error detection: It detects transmission errors using techniques such as the Frame Check Sequence (FCS).
4. Media access control: It determines how devices share and access a common transmission medium.
5. Flow control: Some data-link protocols provide mechanisms to prevent a sender from overwhelming a receiver.
6. Reliable delivery: Some data-link protocols use acknowledgements and retransmissions to improve reliability. This is not provided in the same way by every data-link protocol.
7. Node-to-node delivery: It transfers frames between devices connected through the same link or LAN

Feature	LLC	MAC
Full form	Logical Link Control	Media Access Control
Main purpose	Interface with the network layer	Control access to the medium
Addressing	Not responsible for ordinary Ethernet MAC addressing	Uses MAC addresses
Main Functions	Protocol identification and logical-link services	Framing, addressing and medium access
Example	LLC/SNAP protocol identification	Ethernet and Wi-Fi MAC operations

Question 2: Explain the Ethernet frame format and identify the purpose of destination MAC, source MAC, type/length and FCS fields.

The standard Ethernet MAC frame has a minimum size of 64 bytes and a maximum size of 1518 bytes,

Preamble	SFD	Destination AC Address	Source MAC Address	Type /length	Data and Padding	Frame Check Sequence

Fields and their purposes:

1. Preamble (7 bytes)
The preamble is a sequence used by the receiver to synchronise with the incoming Ethernet transmission.

2. Start Frame Delimiter (SFD) (1 byte)
It marks the end of the preamble and indicates the beginning of the Ethernet frame.

3. Destination MAC Address (6 bytes)
It identifies the intended destination at the data-link layer. Depending on the address, a frame may be intended for one device, a group of devices or all devices on the local LAN.

4. Source MAC Address (6 bytes)
It identifies the transmitting network interface.

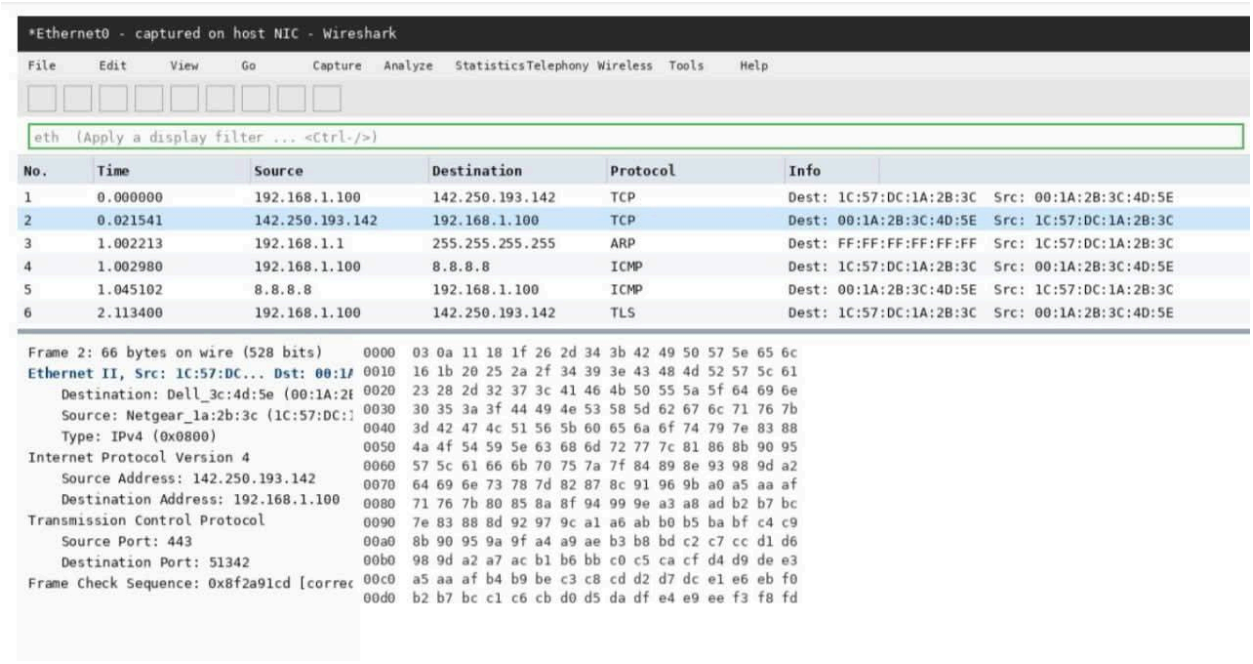
5. Type/Length, (2 bytes)
This field has two common interpretations:
- Values up to 1500 indicate the payload length.
- Values of 1536 (0x0600) and above identify the encapsulated protocol through an EtherType value.
For example, 0x0800 identifies IPv4, while 0x86DD identifies IPv6. The intermediate range is reserved.

6. Data and Padding (46 to 1500 bytes)
This field carries the payload, usually a network-layer packet. Padding is added when necessary to meet the minimum Ethernet frame size.

7. Frame Check Sequence (FCS) (4 bytes)
The FCS contains a 32-bit cyclic redundancy check (CRC) value. The receiver can use it to detect corruption in the received frame. It does not correct errors by itself.

Ethernet frames use MAC addresses for local delivery, the Type/Length field to identify or describe the payload, and the FCS to detect transmission errors.

Question 3: Use Wireshark to capture Ethernet traffic and identify source and destination MAC addresses from at least five frames.



Frame No.	Protocol	Source MAC	Destination MAC	EtherType
1	TCP	00:1A:2B:3C:4D:5E	1C:57:DC:1A:2B:3C	0x0800 (IPv4)
2	TCP	1C:57:DC:1A:2B:3C	00:1A:2B:3C:4D:5E	0x0800 (IPv4)
3	ARP	1C:57:DC:1A:2B:3C	FF:FF:FF:FF:FF:FF	0x0806 (ARP)
4	ICMP	00:1A:2B:3C:4D:5E	1C:57:DC:1A:2B:3C	0x0800 (IPv4)
5	ICMP	1C:57:DC:1A:2B:3C	00:1A:2B:3C:4D:5E	0x0800 (IPv4)

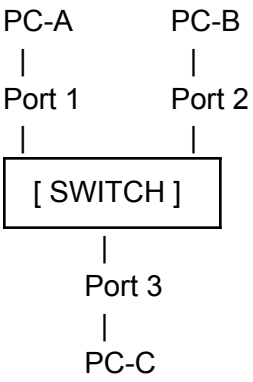
Question 4: Explain the operation of a switch using MAC address learning. Demonstrate how a switch builds its forwarding table.

A network switch is a data-link-layer device that forwards Ethernet frames based on destination MAC addresses. It learns the location of devices by examining the source MAC address of each incoming frame.

MAC address learning process

1. A switch receives a frame on a particular port.
2. It reads the source MAC address.
3. It records the source MAC address and the port on which the frame arrived.
4. It checks its MAC address table for the destination MAC address.
5. If the destination is known, the switch forwards the frame only through the corresponding port, provided that port is eligible for forwarding.
6. If the destination is unknown, the switch floods the frame through eligible ports in the same VLAN, excluding the incoming port.
7. Broadcast frames are also forwarded through eligible ports in the same VLAN, excluding the incoming port.
8. Learned entries are updated when devices move or communicate through different ports, and dynamic entries normally expire after a period of inactivity.

Example network



Assume the following MAC addresses:

- PC-A: AA:AA:AA:AA:AA:AA
- PC-B: BB:BB:BB:BB:BB:BB
- PC-C: CC:CC:CC:CC:CC:CC

Demonstration of MAC learning

Step 1: PC-A sends a frame to PC-B.

The switch learns that PC-A is reachable through Port 1. Since PC-B is not yet in the table, the switch floods the unknown unicast frame through the other eligible ports in the VLAN.

Step 2: PC-B replies to PC-A.

The switch learns that PC-B is reachable through Port 2. It already knows that PC-A is on Port 1, so it forwards the reply only through Port 1.

Step 3: PC-C sends a frame to PC-A.

The switch learns PC-C on Port 3. Since PC-A is known on Port 1, it forwards the frame only to Port 1.

MAC address table

MAC Addres	**Switch Port**	**Learned From**
AA:AA:AA:AA:AA:AA	Port 1	Frame received from PC-A
BB:BB:BB:BB:BB:BB	Port 2	Frame received from PC-B
CC:CC:CC:CC:CC:CC	Port 3	Frame received from PC-C

A switch dynamically builds a MAC address table by learning source addresses and associating them with incoming ports. It then uses this table to forward frames efficiently.

Question 5: Compare hub, switch and bridge in terms of forwarding, collision domains and practical use.

Hub

A hub repeats incoming signals to all other connected ports. It does not inspect destination MAC addresses. All connected devices share the same collision domain, so simultaneous transmissions can collide in a half-duplex Ethernet network.

Bridge

A bridge connects and filters traffic between LAN segments. It learns source MAC addresses and forwards frames based on its forwarding table. It separates collision domains but generally does not separate broadcast domains unless additional Layer 3 or VLAN functionality is used.

Switch

A switch is effectively a multiport bridge. It learns MAC addresses and sends known unicast frames only through the appropriate port. Each switch port creates a separate collision domain; in full-duplex switched Ethernet, collisions do not occur during normal operation.

Feature	Hub	Bridge	Switch
OSI layer	Physical layer (Layer 1)	Data-link layer (Layer 2)	Data-link layer (Layer 2)
Forwarding Method	Repeats incoming signals to other ports	Uses MAC addresses to filter and forward frames|	Uses a MAC address table to forward frames
MAC Address learning	No	Yes	Yes
Collision domains	One shared collision domain	Separates collision domains by segment	Each port is a separate collision domain in traditional Ethernet
Bandwidth	Shared among connected devices	Depends on bridge architecture and links	Dedicated port bandwidth in switched full-duplex Ethernet
Duplex	Typically half-duplex	Depends on implementation	Commonly full-duplex
Practical use	Older, largely obsolete LANs	Connecting or segmenting LAN sections	Modern wired LANs

Question 6: Explain CSMA/CD and discuss why its importance is reduced in modern switched full-duplex Ethernet.

CSMA/CD stands for Carrier Sense Multiple Access with Collision Detection. It is a media-access mechanism used in traditional shared, half-duplex Ethernet.

Operation of CSMA/CD

1. Carrier sensing: A device listens to the medium to check whether another device is transmitting.
2. Multiple access: Multiple devices share the same transmission medium.
3. Transmission: If the medium is idle, the device starts transmitting.
4. Collision detection: While transmitting, the device monitors the medium for a collision.

5. Collision response: If a collision is detected, the device stops transmitting and sends a jam signal.
6. Backoff: The device waits for a random interval determined by the binary exponential backoff algorithm.
7. Retransmission: It attempts to transmit again after the waiting period.

Why CSMA/CD is less important today

Modern Ethernet networks typically use switches with dedicated full-duplex links between each device and the switch.

In full-duplex Ethernet:
- A device can transmit and receive simultaneously.
- Transmissions in opposite directions use separate communication paths.
- There is no shared collision domain on the full-duplex link.
- Collisions do not occur during normal full-duplex operation.
- Therefore, CSMA/CD and its collision-recovery procedure are not required.

CSMA/CD was important in shared half-duplex Ethernet, especially with hubs. Modern switched full-duplex Ethernet largely eliminates the collision problem, making CSMA/CD unnecessary on those links.

Question 7: Explain CSMA/CA in IEEE 802.11 and compare its objective with CSMA/CD.

CSMA/CA stands for Carrier Sense Multiple Access with Collision Avoidance. It is used in IEEE 802.11 wireless LANs to reduce the likelihood of simultaneous transmissions.
Wireless devices may not be able to detect a collision while transmitting as reliably as wired Ethernet devices. Therefore, Wi-Fi uses a collision-avoidance approach.

Operation of CSMA/CA

1. Carrier sensing: A wireless station checks whether the wireless channel is busy.
2. Waiting: If the channel is busy, it waits until the channel becomes available.
3. Interframe spacing: The station waits for the required interframe interval.
4. Random backoff: It selects a random backoff value from a contention window and decreases the counter while the channel remains idle.
5. Transmission: When the counter reaches zero, the station transmits the frame.
6. Acknowledgement: The receiver normally sends an acknowledgement after successful reception.
7. Retransmission: If an acknowledgement is not received, the sender assumes that delivery may have failed, increases contention as required and retries, subject to retry limits.

RTS/CTS mechanism

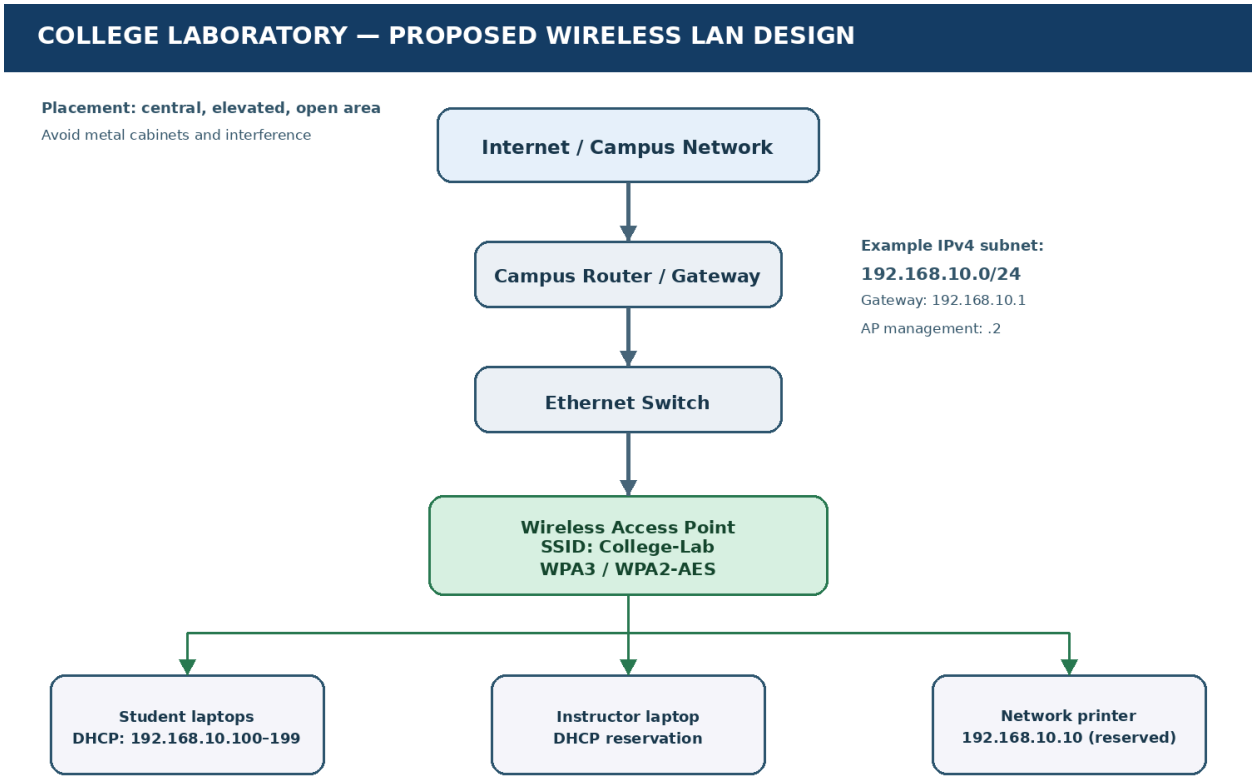
IEEE 802.11 may use Request to Send (RTS) and Clear to Send (CTS) frames to reserve the medium before transmitting data. This can help address the hidden-node problem, where two stations cannot hear each other but can both communicate with the same access point. RTS/CTS is optional and is not used for every frame.

CSMA/CD & CSMA/CA

Both mechanisms allow multiple devices to share a medium. Both use carrier sensing, waiting and backoff to manage access to the medium.

Feature	**CSMA/CD**	**CSMA/CA**
Full form	Collision Detection	Collision Avoidance
Main use	Traditional shared half-duplex Ethernet	IEEE 802.11 wireless LANs
Main objective	Detect collisions after they occur	Reduce the probability of collisions
Transmission medium	Shared wired Ethernet medium	Shared wireless radio channel
Collision handling	Detect, stop, back off and retry	Contend for the channel; infer failures using acknowledgements and retry
Acknowledgements	Not normally used for ordinary Ethernet frame delivery	MAC acknowledgements are commonly used
RTS/CTS	Not part of ordinary Ethernet CSMA/CD	May be used to reserve the wireless medium

Question 8: Design a small wireless LAN for a college laboratory. Specify access point placement, SSID, channel considerations, addressing and basic security requirements.



1. Access point placement

Place the access point near the centre of the laboratory, in an elevated and open position. Avoid metal cabinets and major sources of interference. Conduct a site survey and add managed access points if coverage or capacity is insufficient.

2. SSID

Use College-Lab as the SSID. If required, create a separate staff SSID and a guest SSID with different access permissions.

3. Channel considerations

In the 2.4 GHz band, use 20 MHz channel width and choose a channel based on a survey; channels 1, 6 and 11 are commonly used as non-overlapping choices where permitted. In the 5 GHz band, select a permitted channel with less interference. Consider 6 GHz only if supported by the equipment and local regulations.

4. IP addressing

Device	Example address / configuration
Default gateway	192.168.10.1
Access point management	192.168.10.2
Wi-Fi printer	192.168.10.10
Student laptops	DHCP range: 192.168.10.100–192.168.10.199
Instructor laptop	DHCP reservation or suitable assigned address
Subnet mask	255.255.255.0 (/24)

5. Basic security requirements

• Use WPA3-Personal where supported, or WPA2-AES for compatibility. For a managed college network, WPA2/WPA3-Enterprise with 802.1X and RADIUS is preferable when available.

• Set a strong, unique wireless password and change default administrator credentials.

• Disable WPS if it is not needed and update access-point firmware regularly.

• Use firewall rules and separate student, staff and guest access with VLANs or separate networks where possible.

• Enable guest client isolation when appropriate and restrict access to the access-point management interface.

• Review connected devices and logs periodically.

Expected outcome: The proposed wireless LAN provides convenient connectivity to laboratory devices. Suitable access-point placement improves coverage, channel planning helps reduce interference, DHCP simplifies addressing, and appropriate security controls protect the network.

Question 9: Explain Mobile IP and the need for mobility support when a device moves between networks.

Mobile IP is a networking concept designed to allow a mobile device to move between IP networks while maintaining a stable home IP address for communication. It was designed to

support mobility without requiring every correspondent to know the device's current network location.

Ordinary IP routing relies on IP addresses that identify a device's network location. When a device moves to another network, its point of attachment changes. If it obtains a new IP address, existing sessions may be disrupted unless mobility is supported by the relevant protocols.

- Mobile Node (MN): The device that moves between networks.
- Home Network: The network associated with the mobile node's permanent home address.
- Home Address: The stable IP address assigned to the mobile node in its home network.
- Home Agent (HA): In Mobile IPv4, a router on the home network that maintains information about the mobile node's current location and can tunnel packets to it.
- Care-of Address (CoA): An address representing the mobile node's current point of attachment or reachable location.
- Foreign Network: A network other than the mobile node's home network.
- Correspondent Node: A device communicating with the mobile node.

Working of Mobile IPv4

1. The mobile node starts in its home network and communicates using its home address.
2. When it moves to another network, it obtains or discovers a care-of address.
3. The mobile node registers the care-of address with its home agent.
4. A correspondent node sends packets to the mobile node's home address.
5. The home agent intercepts relevant packets and tunnels them to the care-of address.
6. The mobile node receives the packets through the new network.
7. Depending on the routing arrangement, replies may travel directly to the correspondent node or through a tunnel.

<u>Need for Mobility support</u>

- Continuity of communication: It helps maintain communication as a device changes networks.
- Stable identity: A device can retain its home address while its point of attachment changes.
- Session support: Proper mobility mechanisms can reduce disruption to ongoing applications.
- Location management: The network can maintain information about the mobile node's current reachability.
- Roaming support: It supports movement between networks without relying solely on manual reconfiguration by applications.

Question 10: Perform a packet-level comparison of a wired Ethernet frame and a wireless LAN frame using suitable captures or documented examples.

Wired Ethernet Frame

Destination MAC 6 bytes	Source MAC 6 bytes	Type/Length 2 bytes	Data 46 -1500 bytes	FCS 4 bytes

Wireless LAN Frame(IEEE 802.11 data frame)

Frame Control 2 bytes	Duration/ID 2 bytes
Address 1 6 bytes	Address 2 6 bytes
Address 3 6 bytes	Sequence Control 2 bytes
Frame Body/Payload	FCS 4 bytes

Packet-level analysis

1. Addressing

Ethernet normally contains destination and source MAC addresses in its basic header. An 802.11 data frame can contain more address fields because it may need to identify the transmitting station, receiving station, destination, source and wireless distribution-system relationship.

2. Medium access

Ethernet on a dedicated full-duplex switched link allows simultaneous transmission and reception without normal collisions. Wi-Fi stations share radio spectrum, so they use contention-based channel access and commonly use acknowledgements.

3. Header structure

The basic untagged Ethernet header is 14 bytes, excluding the FCS. A basic three-address 802.11 MAC header is commonly 24 bytes before the frame body and FCS. Optional fields can increase the wireless header length.

4. Error detection

Both Ethernet and ordinary 802.11 MAC frames use an FCS for error detection. In Wi-Fi, failed delivery can also trigger retransmission through MAC-level mechanisms.

5. Frame visibility in Wireshark

An Ethernet capture commonly displays an Ethernet II header. A Wi-Fi capture in monitor mode may display the original 802.11 MAC header, while a capture from a normal Wi-Fi interface may display decapsulated traffic instead.

Field	Ethernet example	Wi-Fi example
Source address	00:11:22:33:44:55	AA:BB:CC:DD:EE:01
Destination address	66:77:88:99:AA:BB	AA:BB:CC:DD:EE:02
Header identification	Ethernet II	IEEE 802.11
Extra information	Type, e.g. IPv4 (0x0800)	Frame type/subtype and sequence number

Comparison table

Feature	Wired Ethernet	IEEE 802.11 Wireless LAN
Standard	IEEE 802.3	IEEE 802.11
Medium	Wired Ethernet link	Wireless radio channel
Basic addressing	Destination and source MAC addresses|	Address fields whose interpretation depends on frame type and direction
Header	Ethernet header, typically 14 bytes without VLAN tagging	Variable MAC header; commonly 24 bytes for a basic data frame without optional fields
Address fields	Destination MAC and source MAC	May include receiver, transmitter, source, destination and BSSID addresses
Frame control	Ethernet has no equivalent 802.11 Frame Control field in its basic header	Frame Control identifies frame type, subtype and control flags

Sequence control	No corresponding general-purpose sequence-control field in the basic Ethernet header	Used in relevant 802.11 frames for sequence and fragmentation information
Error detection	32-bit FCS in the standard frame	32-bit FCS in ordinary 802.11 MAC frames
Access method	Modern switched full-duplex Ethernet does not use CSMA/CD	Uses CSMA/CA for contention-based channel access
Acknowledgement	Ordinary Ethernet does not normally acknowledge every frame at the MAC layer	Unicast frames are commonly acknowledged at the MAC layer, subject to frame type and operating mode

NAME:GRACE ANNA ELSON
CLASS:S5 CSE A
Roll No:33